What is Feasibility study & its type?

ITPM 3 credits

Unit I Project Overview and Project scheduling

Project Overview and Feasibility study

- Project overview → for 2M

means → software, system, complete summary of project.

Project → complete summary →

↓

Project = ?

cause = ?

Benefit = ?

i) Project Name
ii) Project discription. → details of project
iii) objective → what is reason
iv) Scope → users of the project
v) Expected outcome. → Feedback

- Feasibility study.

Project → ready → Possibility ✗

Profit ✗

eg. college

ERP – Budject = ?

Developers = ?

Time = ?

s/w execute = ✓✗

There are 4 type of feasibility study.

(1) Technical Feasibility
Prj - Technology ✓x
H/w = ?
s/w = ?
Internet study = ?
Proj. lang = ?
Database = ?

(2) Economic feasibi
Proje - Profit ✓x
Project cost = 10k
Projit = ?

(3) Operational Feasibility
user - system use ✓x
staff - work ✓x
staff - Training ✓x
Users - happy ✓x

(4) Schedule Feasibilit
Project - Time ✓x
eg. Decide - 3M
complete - 3M ✓x

What is Identification & its step.

Identification → find the problems

↓

According to problem

↓

create the s/w

↓

To solve problem.

Problem = ?

s/w ✓ ✗

Check there is need to create the s/w or not

1. Identify the problem. → ex No online Atten System
2. Analyze current situation → ex Physical Atten Takes m
3. Identify stakeholders → ex Teachers, students e
4. Gather req. → Take interview, survey, Mettings to ch (chief use techniques)
5. Define prj object → set the objective (target).
6. Identify possible solution. → Mob A, Web Applicat
7. Preliminary feasibility check → T, S, O, E fe
8. Prepare project proposal → There is need of project or not

Que Explain the concept of Market & demand Analysis

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Market and demand Analysis.

Definition

Before starting / II project → studying the market
customers competitors how many competitors
future demand for a product. → Product's future demand

Market Analysis

Overall Market Environment → study
study of entire market

- Market growth
- Industry trends
- Target customers
- Pricing strategy

eg. Food delivery App - No of online users
market trends
food delivery services.

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Demand Analysis

How many customer? → buy = ? / use = ? } Product.

- customer needs.
- Product price.
- Population.
- seasonal demand.
- Income level.

eg. online learning platform - develop.

- No of students.
- Internet availability.
- Demant for online course.

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Steps to perform Market & Demand Analysis

1. Identify the target market.
   - who will use the product = ?
2. collect market Information.
   - surveys, Interviews, online search.
3. Analyze customer needs.
   - customer expectations.
   - Pain points.
   - Desired features.
4. Analyze competitors.
   - Product offered
   - Pricing = ?
   - weakness = ?
5. Estimate Market Demand.
   - No. of Potential
   - Future growth = ?
   - Expected sales.
6. Forecast Future demand.
   - Prediction. - Future demand.
   - Historical data.
   - F/B of customer

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⑦. Prepare report.
- Market size ?
- Target.
- Risks.
- Business opportunities.

Project Cost Estimation.

Process of → predicting the total cost req. to complete an IT project.
- cost of resources
- labor
- H/w, s/w cost.
- Training
- Testing
- Maintenance.

objective - Project Budjet → overall project ka budjet find. karna.

Objectives

① Determine the total project budjet
② supports project planning & scheduling.
③ Allocate resources.
④ Reduce financial Risks.

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Qu Explain the concept of Project Cost estimate along with components & objectives

components of project cost estimate.

① Human resource cost.
- project management.
- s/w dev.
- Tester.
- UI/UX design
- Business Analyst

② H/w cost.
- cost of physical equipment.
- computer.
- servers.
- N/w devices
- Printers

③ s/w cost.
- Expenses related to s/w licenses.
- OS.
- Databases.
- Antiviruses

④ Training cost.
- Training emp on new technologies
- Workshops.
- Instructor fees.
- Online certificates.

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(5) Infrastructure cost

~~- cost of physical equipment~~
~~- computer.~~
~~- server.~~
~~- N/w devices.~~
~~- printer.~~

- Internet.
- Electricity
- office rent
- furniture.

Financial Appraisal

→ Process - check, s/w - profit = ✓
feasible = ✓.

→ check - Project reason - evaluate.

objectives

(1) To estimate the total proj cost.
(2) To estimate profit.
(3) To compare diff investors.
(4) To minimize financial risks.

Components

(1) Initial Investments.
Amt of money → start the project
(2) H/w cost.
(3) s/w cost.
(4) office setup.
(5) Emp training.

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(2) operating cost.

- Repeated expenses. — during project.

- Emp salary.
- Electricity bills
- Maintainance.
- Internet charge

(3) Revenue Estimation.

expected income → generated from project.

- Product sales.
- service charges.

(4) cash flow Analysis.

- Movement of money IN/OUT.
- cash Inflow (Income) ★★
- cash outflow (expenses) ★★

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Financial Appraisal Techniques.

(1) Payback Period. kitne salo me fhydd hoga

$$PBP = \frac{\text{Initial Investment}}{\text{Annual cash inflow}}$$

(2) Return on Investment (ROI).

$$ROI = \frac{\text{Net profit}}{\text{Investment}} \times 100.$$

(3) Benefit cost Ratio.

$$BCR = \frac{\text{Percent value of Benefit}}{\text{percent value of cost}}$$

Advantages

(1) Helps in Investment decision.
(2) Measure profit.
(3) Reduce risk (finance)

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Project scheduling.

Process → crating (TT) → time table
for project
Activities

- sequence of task
- duration = ?
- assigning resources = ?
- decision of start & end date.

Project activites ka time table create karna.

objectives :

(1) planned deadline - project complete on time.

(2) resources proper use.

(3) Project progress monitor

(4) Project risk ↓

Steps in project schedulling

(1) Define project activities.

dividing the project in small activities

Project → □ □ □ - act = ?
-11-
-11-

(2) sequnce activites - determine order.
arranges the activities in order

(3) Estimate Activity Duration.

(4) Assign Resources
- Allocate Team member
- H/w
- S/W

(5) Develop the project schedule
- schedule - Gantt chart, pert, CPM.
critical path Metho

(6) Monitor and update.
Track
Progress - regularly.

Benefits

(1) Timely completion of projects.

(2) Better Monitoring.

(3) Improves project planning.

Critical Path

Critical Path is the detective path of the activities which takes minimum amount of time in order to perform it.

start with jo activity kisi pe dependent nahi hai. usse start hoga.

ES - Early start

EF - Early Finish

D - Duration

LS - Late start

LF - Late Finish

SL - Slak value.

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EF = EST + DUR.

SL = LF − EF

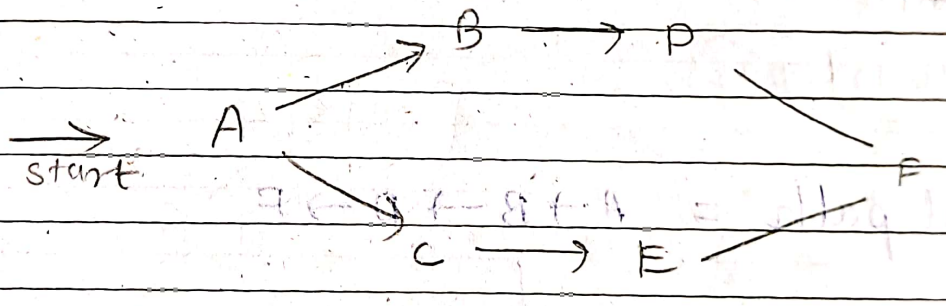
LS = LF − D.

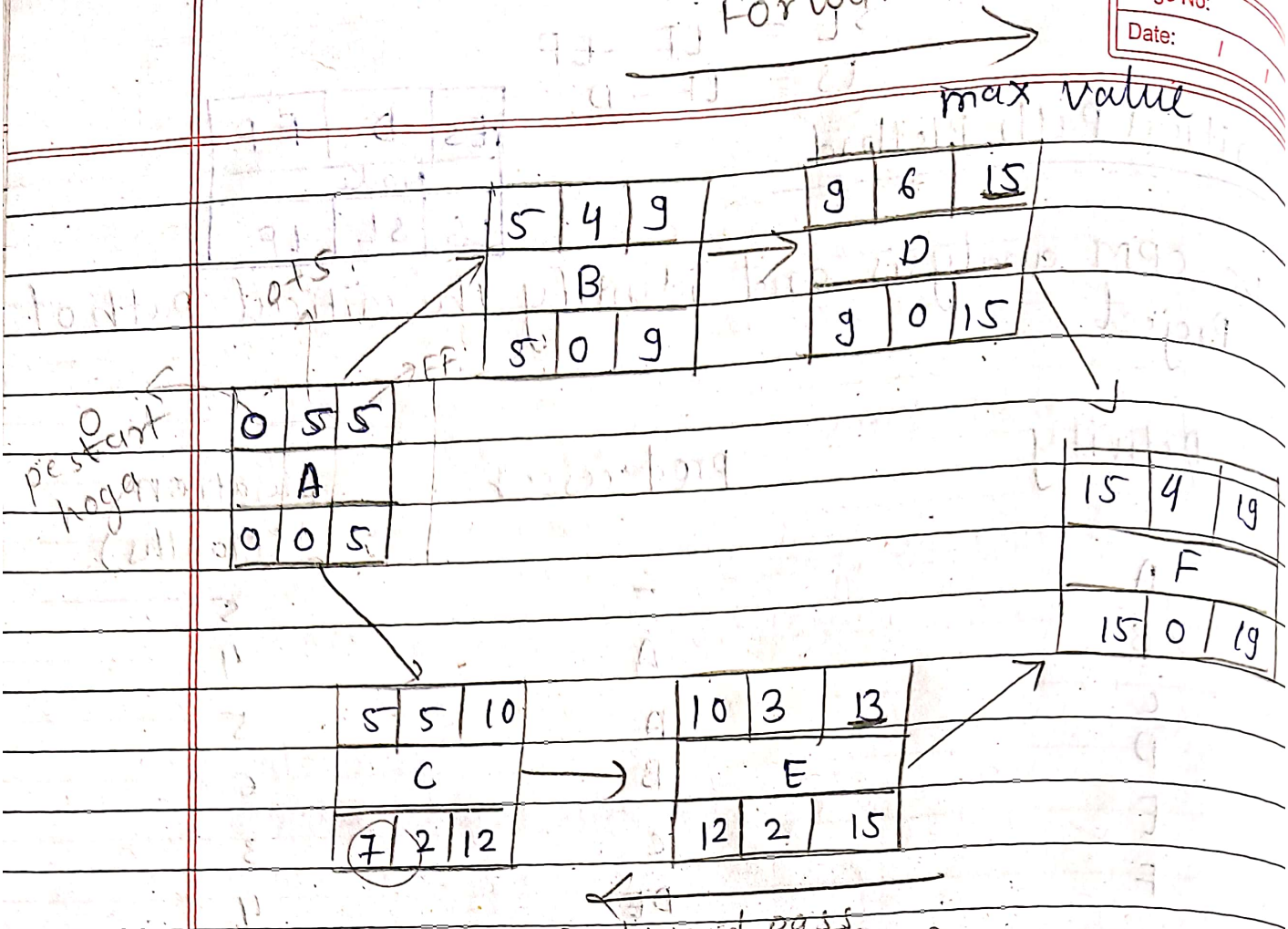
ES	D	EF
Node		
LS	SL	LF

Critical Path Method.

Q. Do CPM analysis and identify the critical path of Project.

Activity	Predecessor	Duration (Months)
A	–	5
B	A	4
C	A	5
D	B	6
E	C	3
F	DE	4





min value.

↓

consider min value for backward pass.

15 & 13 me se max value li = 15

critical path = A → B → D → F

↓

we choose that critical path which has slack value = 0.

Pert
Float

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5M

Difference between Pert and CPM.

	PERT	CPM
Nature	Probabilistic	Deterministic
Time Estimath.	User 3 times 1) optimistic (t_o) 2) Most likely (t_m) 3) ~~Probabilistic~~ pessimistic (t_p)	single time Duration.
Focus	Time management	cost control.
Suitable for	Development	Manufacturing repetative project.
orientation	Event	Activity
Cost consideration	Time - importance cost - x.	cost → importan
Expected Time	$T_e = \frac{t_o + 4t_m + t_p}{6}$	No formula
critical path	vary ↓ vary due to sitcath change	fixed.

2M Precedence Relationship.

There are 4 type of PR

	Name	Abrevation	work
①	Finish to start(2)	F S	Finish 1st Activ before starting the next
②	start to start	S S	start both activitie together
③	Finish to finish.	F F	Both Activities must finish togt
④	start to Finish.	S F	First Activity mu start before secor can finish

Que what is crashing & how crashing playing important role in PM. for cost reduction in CP metho

Cost Reduction by crashing of Activities

crashing: Tech - PM - used to complete project faster by adding extra resources to critical activities.

key points:

1) ↓ proj completion time.
2) Focuses on critical path activities.
3) ↑ direct cost (extra staff, overtime)
4) ↓ Indirect cost (administration, office expenses.

steps:

① Identify CP
② Find activities crashed.
③ calculate cost.
④ crash the least expensive critical activity
⑤ Repeat until the target time is achived.

Adv:

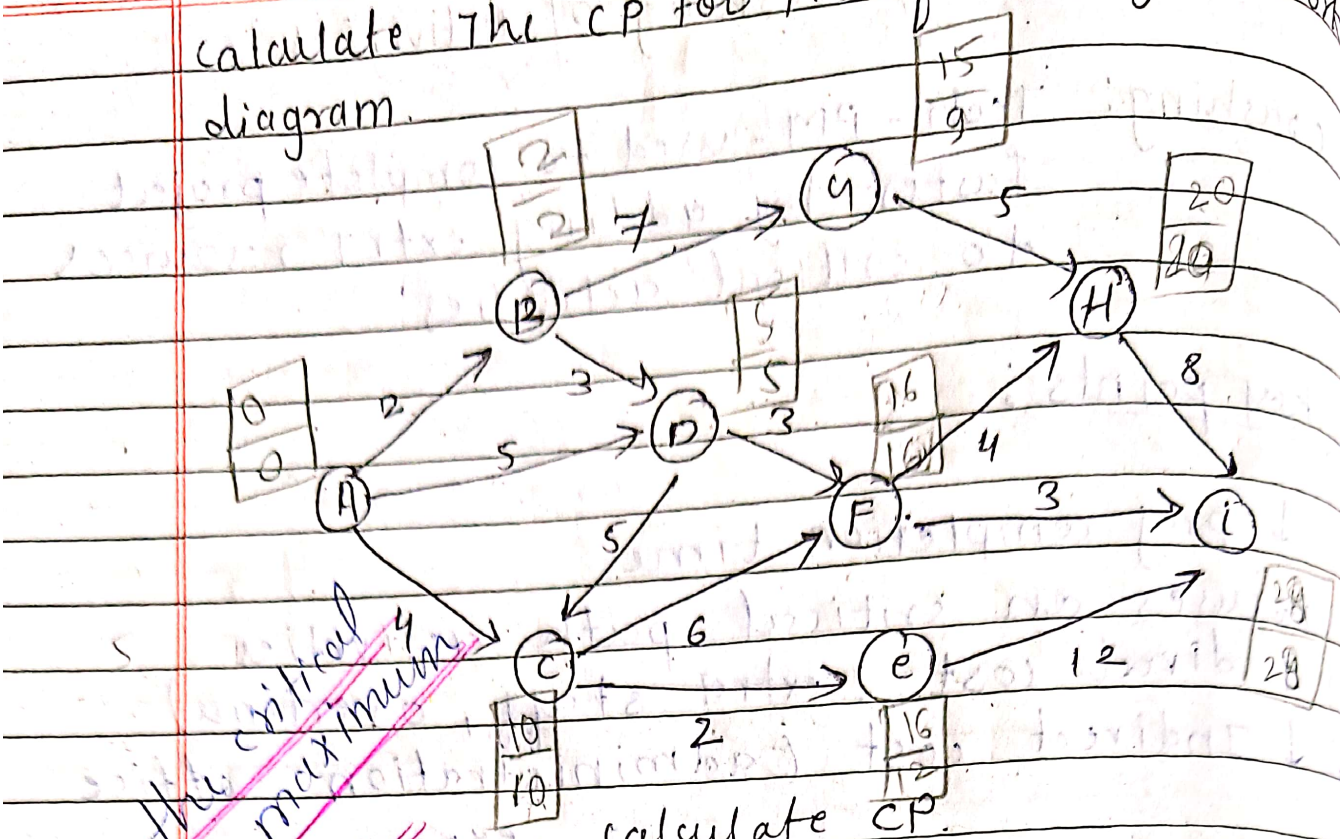
① Finishes the project earlier.
② Reduces overall cost.
③ Helps meet deadlines.

Dis.

① ↑ Direct cost
② Not all activities can be crashed.

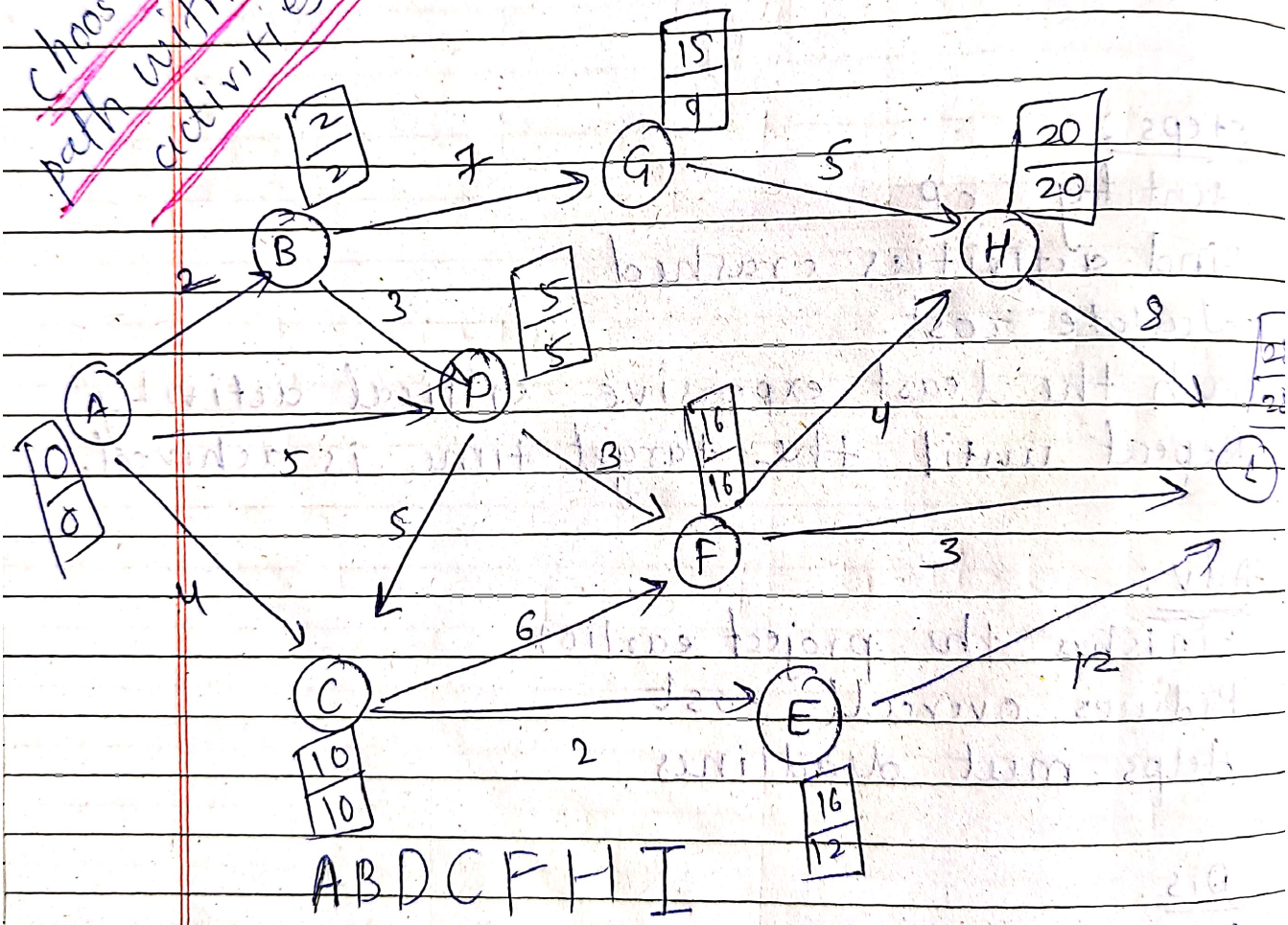
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calculate the CP for the following Network diagram.



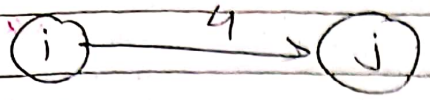
calculate CP.

Choose the critical path with maximum activities.



ABDCFHI

Float calculation.

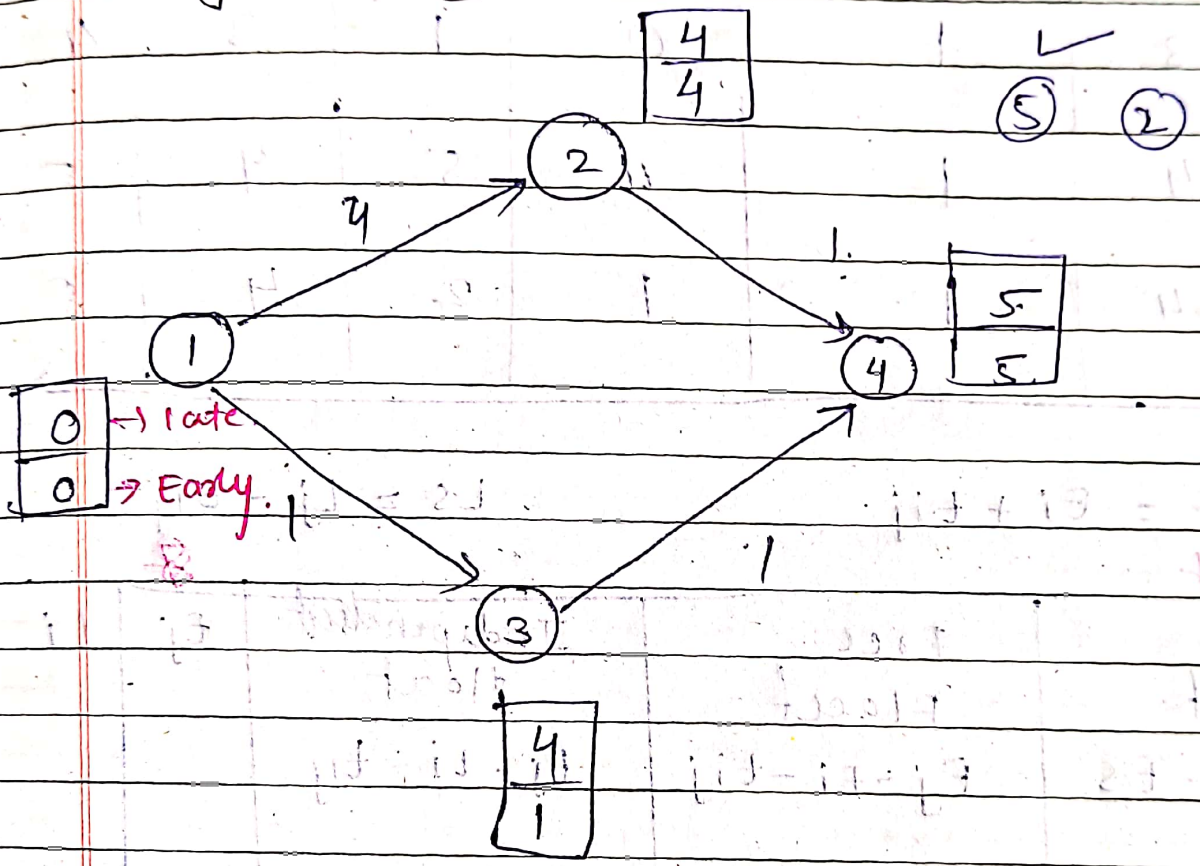


Tij = Total duration.

Esij → (1) → start
EFij → (1) → Finish
Lsij → (2) → start
LFij → (2) → Finish

} Early = (1) activity

} late = (2) activity



Float = criticcal path.

F = latest − Earliest.

Que. consider the ... pass & Backword pass, also ... in the term of total float, independ; free float

Float → Total Float
→ independent float
→ free float

Ei → 1st node ka early start

Lj → second node ka latest

sequence to fill the colums

Activity	Normal time (tij) tij	Early start Es (Ei)	Early finish Ef	Latest start Ls	Latest Finish Lf(Lj)
	2	3	5	6	7
1					
1-2	4	0	4	0	4
1-3	1	0	1	3	4
2-4	1	4	5	4	5
3-4	1	1	2	4	5

EF = Ei + tij

LS = Lj - tij

Total Float = LS - ES	Free Float Ej - Ei - tij	Independent float Ej - Li - tij	Ej	Li
0	0	0	4	0
3	0	0	1	0
0	0	0	5	4
3	3	0	5	4

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E_j = early start of last node.

L_j = late start of first node

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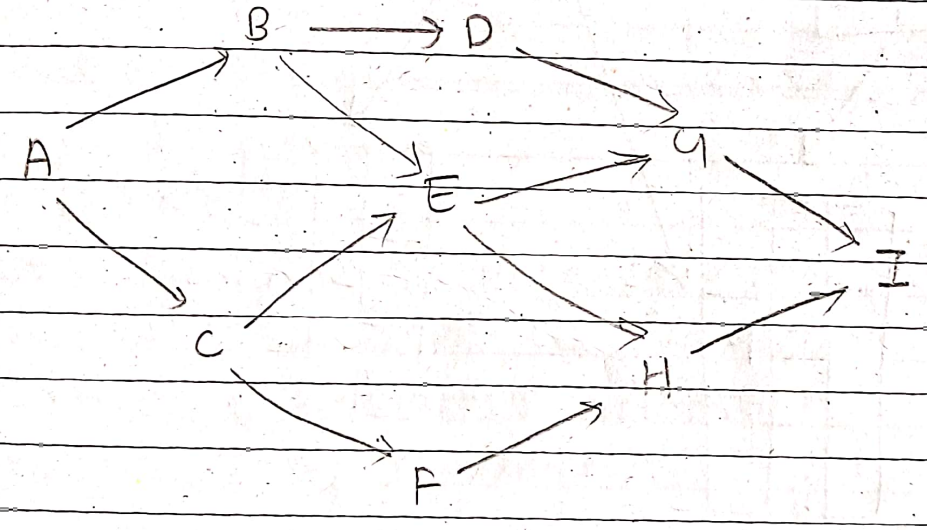
Total float jaha zero $\xrightarrow{\text{V0}}$ critical path.

Imp

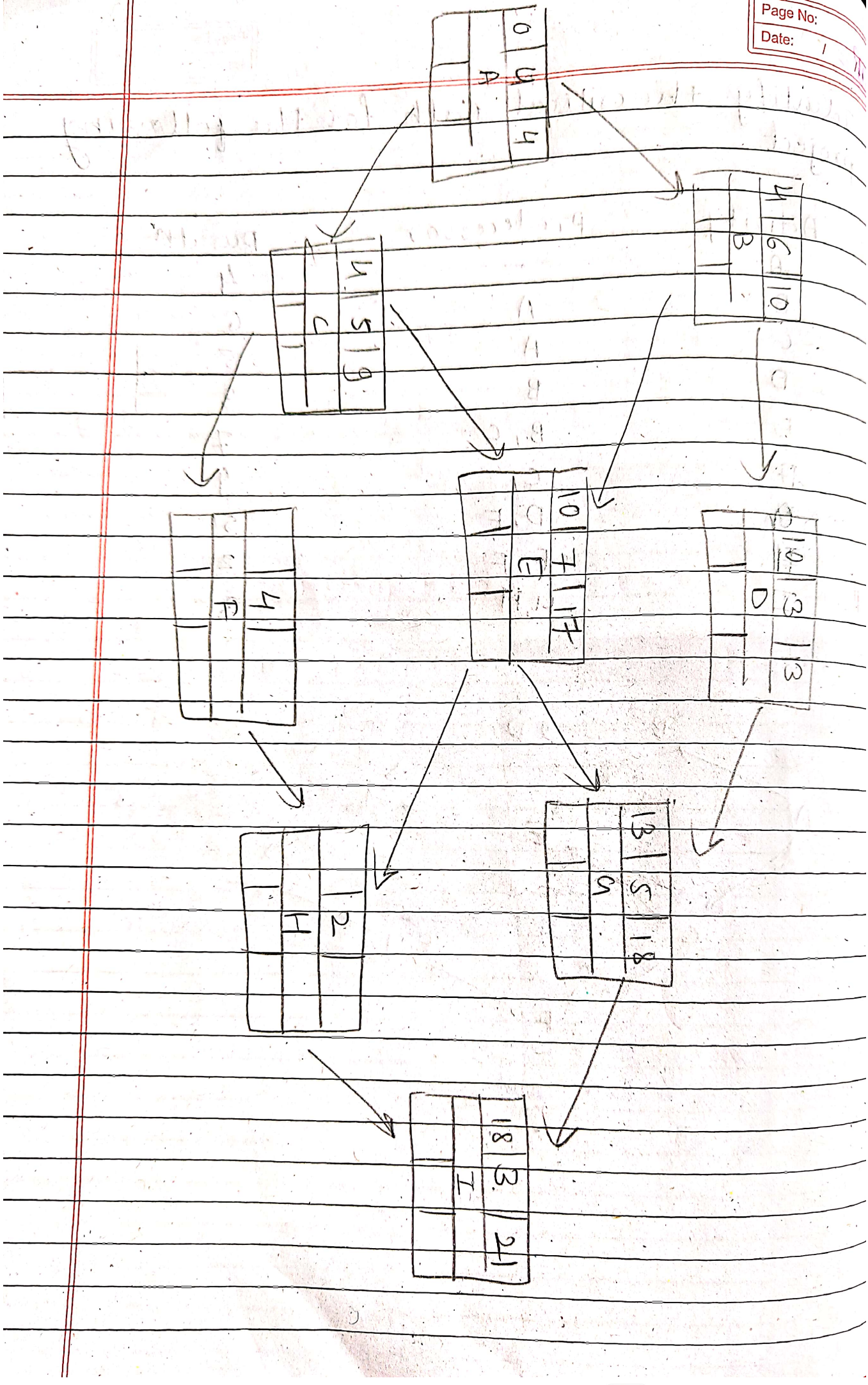
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Q. Identify the critical path for the following project.

Activity	Predecessor	Duration
A	—	4
B	A	6
C	A	5
D	B	3
E	B, C	7
F	C	4
G	D, E	5
H	E, F	2
I	G, H	3



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A
0
C
4
C
4
5
6
B
4
6
10
E
7
11
D
10
3
13
F
4
1
H
1
2
S
13
5
18
T
18
3
21

PERT calculation.

Activity	Estimated duration		
	to optimistic	tm most likely	tp pessimistic
1-2	1	7	13
1-6	2	5	14
2-3	2	14	26
2-4	2	5	8
3-5	7	10	19
4-5	5	5	17
6-7	5	8	29
5-8	3	13	9
7-8	8	17	32

calculate → Based on Activity

1) Draw project N/W diagram.
2) Expected duration & varience
3) Earliest and latest accurance
4) Expected project length.

Formula: expected time

$$te = \frac{to + 4tm + tp}{6}$$

varience:

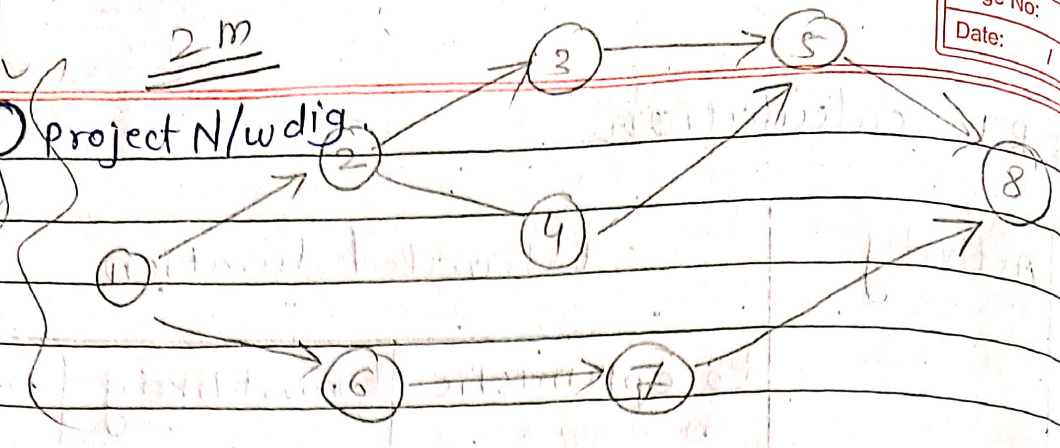
$$\sigma = \left(\frac{tp - to}{6}\right)^2$$

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Create the diagram based on Activity

2 m

(1) Project N/w dig.



(2) Expected duratn & variance.

(1) 1-2	(2) 1-6
$te = \frac{to + 4tm + tp}{6}$	$te = \frac{2 + 4\times5 + 14}{6}$
$= \frac{1 + 4\times7 + 13}{6}$	$= \frac{2 + 20 + 14}{6}$
$= \frac{1 + 28 + 13}{6}$	$= \frac{36}{6}$
$= \frac{42}{6}$	$= 6$
$= 7.$	
$\sigma = \left(\frac{13 - 1}{6}\right)^2$	$\sigma = \left(\frac{14 - 2}{6}\right)^2$
$= \left(\frac{12}{6}\right)^2$	$= \left(\frac{12}{6}\right)^2$
$= 4.$	$= 4.$

1 28 13 1 2

22 14

1
28
56
84

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(3) 2 - 3

$$te = \frac{to + 4tm + tp}{6}$$

$$= \frac{2 + 4 \times 14 + 26}{6}$$

$$= \frac{2 + 56 + 26}{6}$$

$$= \frac{84}{6}$$

$$= 14$$

$$6 = \left(\frac{26 - 2}{6}\right)^2$$

$$= \left(\frac{24}{6}\right)^2$$

$$= 16$$

(4) 2 - 4

$$te = \frac{2 + 5 \times 4 + 8}{6}$$

$$= \frac{2 + 20 + 8}{6}$$

$$= \frac{22 + 8}{6}$$

$$= \frac{30}{6}$$

$$= 5$$

$$6 = \left(\frac{8 - 2}{6}\right)^2$$

$$= \left(\frac{6}{6}\right)^2$$

(5) 3 - 5

$$te = \frac{7 + 4 \times 10 + 19}{6}$$

$$= \frac{7 + 40 + 19}{6}$$

$$= \frac{66}{6}$$

$$= 11$$

$$6 = \left(\frac{19 - 7}{6}\right)$$

$$= \left(\frac{12}{6}\right)^2 = 4$$

1
47
9
66

(6) 4 - 5

$$te = \frac{5 + 4 \times 5 + 17}{6}$$

$$= \frac{5 + 20 + 17}{6}$$

$$= \frac{25 + 17}{6} = \frac{42}{6} = 7$$

$$6 = \left(\frac{17 - 5}{6}\right) = \left(\frac{12}{6}\right)^2 =$$

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(7) 6-7.

$te = \frac{5 + 4x8 + 29}{6}$

$= \frac{5 + 32 + 29}{6}$

$= \frac{37 + 29}{6}$

$= \frac{66}{6} = 11.$

$6 = \left(\frac{29 - 5}{6}\right)^2$

$= \left(\frac{24}{6}\right)^2 = 16.$

(8) 5-8

$te = \frac{3 + 4x3 + 9}{6}$

$= \frac{3 + 12 + 9}{6}$

$= \frac{15 + 9}{6} = \frac{24}{6} =$

$6 = \left(\frac{9 - 3}{6}\right) = \frac{6}{6} = 1$

(9) 7-8.

$te = \frac{8 + 4x17 + 32}{6}$

$= \frac{8 + 68 + 32}{6}$

$= \frac{8 + 100}{6}$

$= \frac{108}{6}$

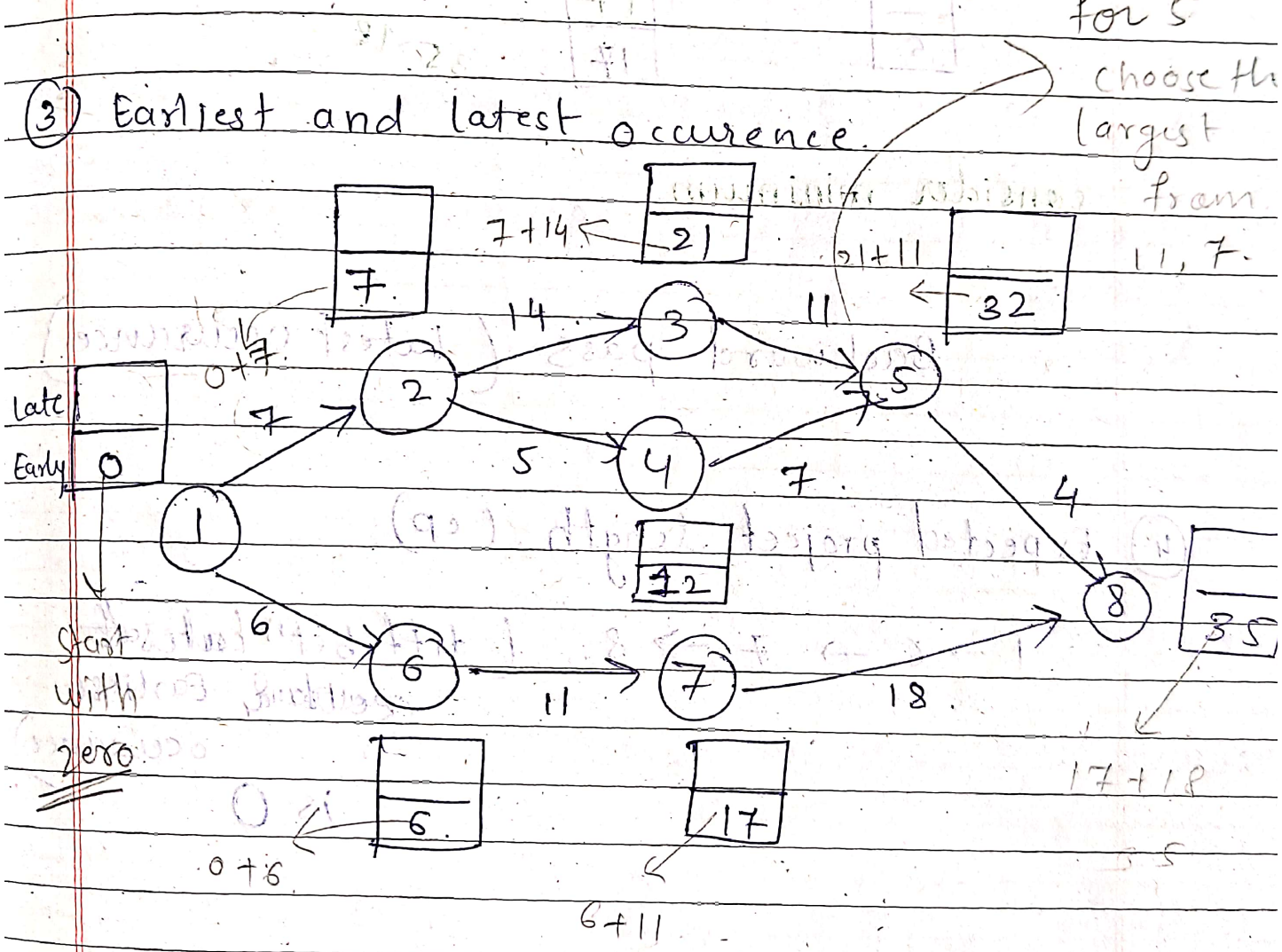
$= 18.$

$6 = \left(\frac{32 - 8}{6}\right)^2.$

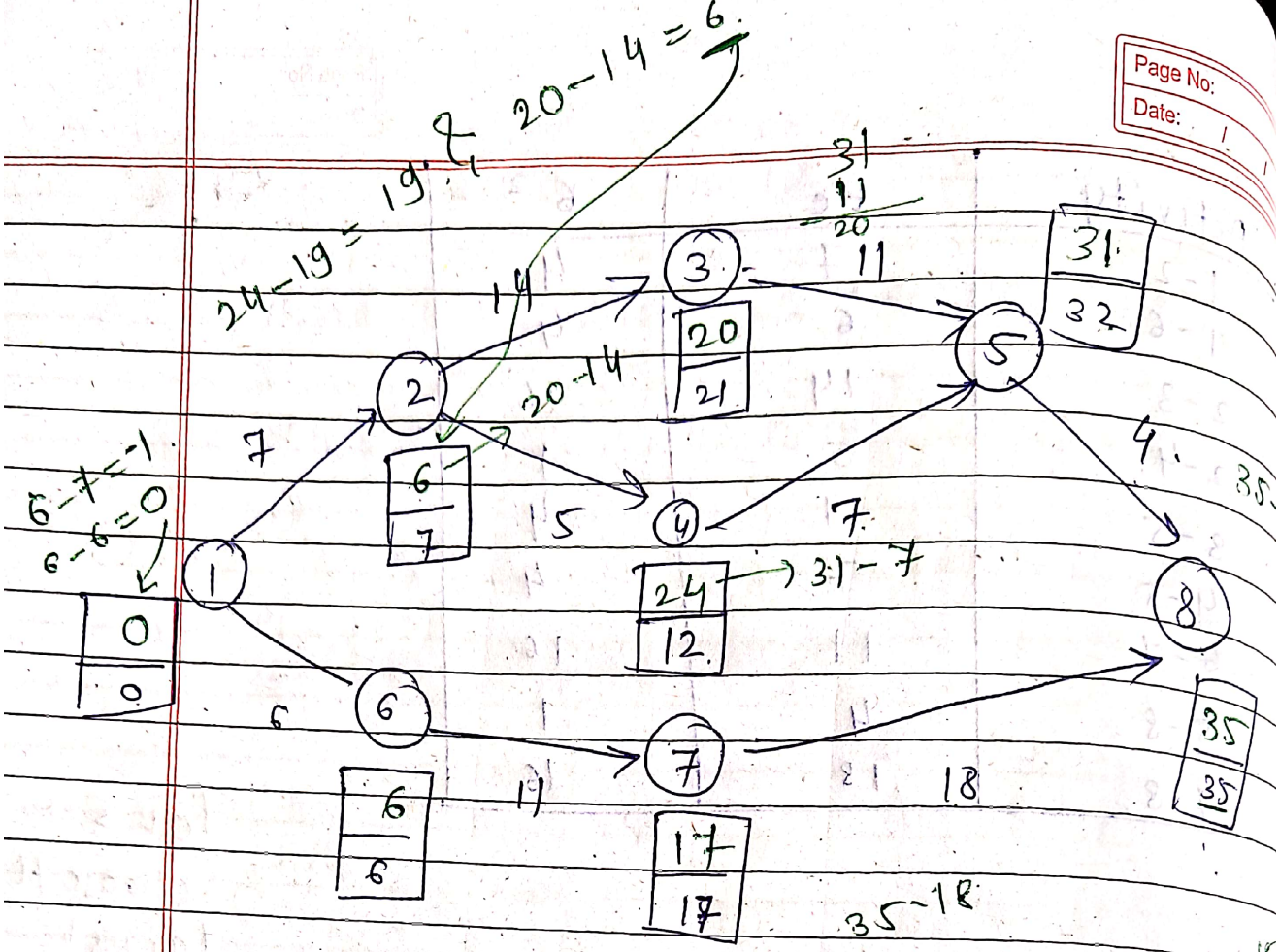
$= 16.$

Activity	te	σ^2
1-2	7	4
1-6	6	4
2-3	14	16
2-4	5	1
3-5	11	4
4-5	7	4
6-7	11	16
5-8	4	1
7-8	18	16

(3) Earliest and latest occurence.



forward pass (Earliest occurance)



consider minimum.

Backward pass (latest occurance)

(4) Expected project length (cp)

1 → 6 → 7 → 8. (diff bet latest occurance and Earliest occurance is 0